LOVELY PROFESSIONAL UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

# INT428: ARTIFICIAL INTELLIGENCE FOUNDATIONS

## CAPSTONE PROJECT OFFICIAL SUBMISSION DOCUMENT

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| **Course Code & Title:** INT428: Artificial Intelligence Foundations | **Group Number:** Group 13 |
| **Official Allocated Topic:** Customer Support Chatbot & Operational Management Portal for a Real Small Business | **Candidate Name:** Suman Saha |
| **Target Enterprise / Location:** Razors-n-scissors (BH-5, LPU Campus) | **Registration Number:** 12524973 |

## PART 1: PROJECT BRIEF & BASIC DETAILS

**Official Allocated Title:** *Customer Support Chatbot & Operational Management Portal for a Real Small Business*

This project implements an autonomous conversational AI assistant and real-time operations management system custom-built for **Razors-n-scissors** (BH-5 Boys Hostel, Lovely Professional University Campus). The solution eliminates front-desk telephone interruptions and prevents missed after-hours reservations by automating 24/7 appointment scheduling, service inquiries, grievance registration, and escalation queues directly over official WhatsApp without human staff intervention.

**Key Technical Specifications & Syllabus Implementations:**

- **Dual AI Engine (INT428 Syllabus Requirement):** Implemented and benchmarked *Approach 1 (Deterministic Rule-Based FSM Slot-Filler)* against *Approach 2 (Dense Sentence-Transformers Semantic Retrieval)*, unified via a production Hybrid Arbiter with live explainability traces.
- **Multi-Turn Appointment Scheduling:** Direct one-click service selection, multi-service combo support (e.g., Hair Cut + Beard Trim with automatic duration aggregation and package discount), and live date/time slot filling.
- **Strict Interval Overlap Prevention:** Mathematical chair conflict prevention ensuring no two clients are booked simultaneously on the same seat using interval intersection logic ($start_{\blacksquare} < end_{\blacksquare} \wedge end_{\blacksquare} > start_{\blacksquare}$).
- **24/7 Cloud Architecture (Zero Laptop Dependency):** Deployed on Render Cloud with persistent PostgreSQL database preserving multi-device session keys permanently across restarts, synced in real-time with a Streamlit Cloud operations dashboard.
- **Responsible AI Guardrails:** Automated DPDP Act PII masking in audit logs and clinical boundary guardrails refusing medical scalp advice.

## PART 2: INDIVIDUAL CONTRIBUTION (GROUP 13)

**Statement of 100% Sole Individual Authorship:**
Although Group 13 was officially allocated with three roster slots on university records, the entire capstone project was **solely conceptualized, architected, coded, benchmarked, tested, and deployed individually by Suman Saha (Registration No: 12524973)**.

The other two allocated group members did not communicate, coordinate, write code, contribute documentation, or participate in any phase of this project. Every technical artifact—including the dual AI models, relational database schemas, real WhatsApp WebSocket gateway, Streamlit live counter dashboard, automated 19-test test suite, and cloud hosting infrastructure—was executed solely by Suman Saha.

## PART 3: PUBLIC GITHUB REPOSITORY LINK

In accordance with official submission requirements, the complete open-source repository containing source code, test suites, architecture schemas, and deployment scripts is available at the link below:

**Public GitHub Repository Link:** https://github.com/0x-ryo/Chatbot